



SAKSHAM GUPTA

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Location: New Delhi, Delhi

EDUCATION

Manipal Institute of Technology, Manipal *Manipal, Karnataka • May 2027*
B.Tech in Information Technology (CGPA: 9.15/10.0)
Delhi Public School R.K. Puram *New Delhi • 2023*
Class 12 – CBSE (90.2%)
Class 10 – CBSE (93.5%)

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C++
Computer Science Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems
Software Engineering: SDLC, Agile Methodology, Requirements Engineering, Software Testing
Databases, Tools & Security: SQL, MongoDB, Firebase, VS Code, Git, Docker
Web & Backend Development: Node.js, Express.js, React.js, Flask, REST APIs, Socket.io, JWT Authentication, AWS
Artificial Intelligence & Machine Learning: Pandas, NumPy, Scikit-learn, LLM Integration, RAG Pipelines, NLP

ACHIEVEMENTS & CERTIFICATIONS

- **Achievers Scholarship** - Ranked in Top 5% of Information Technology cohort at MIT Manipal
- **Academic Excellence** - Maintained consistent CGPA of 9.15/10 across all semesters
- **AWS Generative AI Applications** - Professional Certificate (Coursera)
- **Google Cloud Digital Leader** - Professional Certificate (Coursera)

EXPERIENCE

Software Development Intern *June – July, 2025*
Indian Railway Catering and Tourism Corporation (IRCTC)

- Analyzed enterprise booking system (2M+ daily transactions), mapping data flow and identifying observability gaps and performance bottlenecks
- Optimized relational database queries by 25% via execution plan analysis and index tuning; documented insights for monitoring
- Evaluated microservices and distributed tracing for scalability; automated load tests using Python to simulate concurrent users
- Built Python scripts to monitor query performance and simulate concurrent workloads, improving system reliability

PROJECTS

SafePay | ML-Based Transaction Fraud Detection System | [GitHub](#) • [View Live](#)
Python, Flask, MongoDB, Scikit-learn, REST API, Cryptography

- Built an ML fraud detection model with feature engineering achieving 85%+ accuracy; exposed predictions via a Flask REST API for real-time transaction validation
- Integrated model inference pipeline with MongoDB backend, enabling low-latency decision scoring on incoming transaction payloads
- Applied end-to-end security using AES-256 encryption, digital signatures, and secure password hashing

TripGenie | AI-Powered Role-Based Travel Planning Platform | [GitHub](#) • [View Live](#)
React, TypeScript, Firebase (Auth, Firestore, Storage), Gemini API, Tailwind CSS, Vite

- Built a role-based travel platform (Traveller, Agent, Admin) using React + TypeScript with Firebase for auth, real-time data, and cloud storage
- Built a RAG pipeline with semantic retrieval and generation optimization using Gemini API, improving response relevance and reducing hallucinations
- Designed a structured Firestore data model supporting plan discovery, itinerary management, and admin oversight across roles

Exoplanet Habitability Analysis & Discovery Dashboard | [GitHub](#) • [View Live](#)
Python, Pandas, NumPy, Scikit-learn, PCA, K-Means, React, Vite, Tailwind CSS

- Built a data mining pipeline using NASA Exoplanet Archive data with KNN imputation, PCA, and clustering to identify Earth-like exoplanets
- Developed a habitability scoring model and React dashboard with similarity search and live prediction for exoplanet candidates
- Visualized planetary clusters and feature relationships using dimensionality reduction and correlation analysis against habitable-zone candidates

AgriSmart | ML-Based Crop Recommendation & Revenue Predictor | [GitHub](#) • [View Live](#)
Python, Flask, Scikit-learn, Pandas, NumPy, JavaScript, Leaflet.js, Cloud

- Trained a stacking classifier (Random Forest, Gradient Boosting, AdaBoost) on crop, rainfall, and market data achieving 90% classification accuracy
- Paired with a Gradient Boosting regressor for revenue-per-hectare estimation; exposed both models via a Flask REST API for real-time inference
- Built an interactive district-level India map (Leaflet.js + GeoJSON) for live predictions, deployed on Vercel

LEADERSHIP & ACTIVITIES

Organizing Committee Member — Sports (Chess) *March 2025*
Revels'25, MIT Manipal

- Led logistics coordination for chess tournaments with 200+ participants, managing registration and bracket systems
- Collaborated with a 12-member cross-functional team ensuring 100% on-time event execution during cultural fest